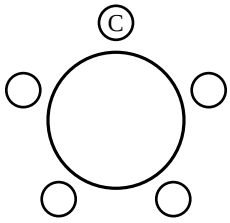
Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 38295

Part A: Each question is worth 3 points

1. Five friends, Clara, Emma, John, Anna and Ben, go to a restaurant and sit at a table with 5 chairs. Clara always sits on the same chair (marked C).



In how many different ways can the friends sit around the table?

A. 24 ***B.*** 16 ***C.*** 8 ***D.*** 5 ***E.*** 4

2. The digits from 1 to 9 are written on 9 cards. Mia has 5, 3 and 2; Tom has 8, 7 and 6; Fred has 9, 4 and 1. Each of them uses some of +, –, ×, ÷ (and brackets) and each of their own cards exactly once.

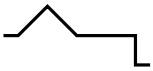
Who *cannot* get 40 as the result?

A. Mia ***B.*** Tom ***C.*** Fred

D. All can get 40 ***E.*** Mia and Tom

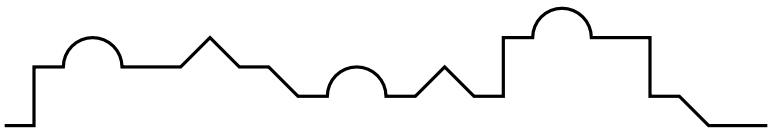
3. Lena cut a strip of paper along this line.

Which piece **cannot** come from the cut line?

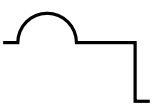
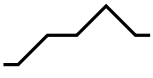
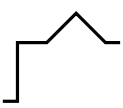
A. ***B.*** ***C.*** ***D.***  ***E.***

Part B: Each question is worth 4 points

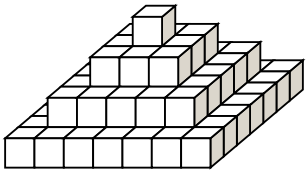
4. This is the outline of a town seen from the river at sunset.



Which piece **can** be part of the outline?

A. ***B.*** ***C.***  ***D.***  ***E.*** 

5. Sara built this pyramid out of identical cubes. Every layer is a full square, with no holes hidden inside.



How many cubes are in the pyramid?

A. 16 ***B.*** 42 ***C.*** 58 ***D.*** 83 ***E.*** 84

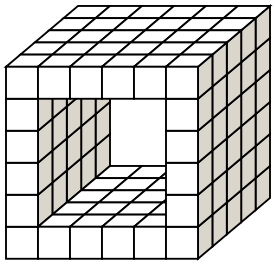
6. Dan has a rope 6 m long and cuts it into pieces that are each 40 cm long.
How many pieces does Dan get?

A. 150 ***B.*** 30 ***C.*** 16 ***D.*** 15 ***E.*** 14

7. Sara builds a brick 6 cubes long and 4 cubes tall from red and blue cubes. The outside is completely red and the inside is completely blue. Sara used exactly 8 blue cubes.
How many cubes deep is the brick?

A. 5 ***B.*** 4 ***C.*** 3 ***D.*** 2 ***E.*** 1

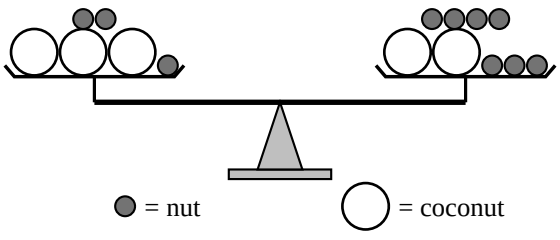
8. Sara built this tunnel out of identical cubes. The hole goes straight through from the front to the back. Sara wants to fill the hole completely, so that the tunnel becomes one solid block.



How many more cubes are needed?

A. 80 ***B.*** 68 ***C.*** 64 ***D.*** 20 ***E.*** 16

9. The balance is level: 3 coconuts and 3 nuts on one plate, 2 coconuts and 7 nuts on the other.

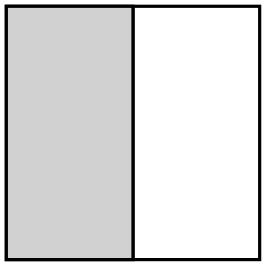


One coconut weighs the same as how many nuts?

A. 1 nut ***B.*** 2 nuts ***C.*** 3 nuts ***D.*** 4 nuts ***E.*** 7 nuts

Part C: Each question is worth 5 points

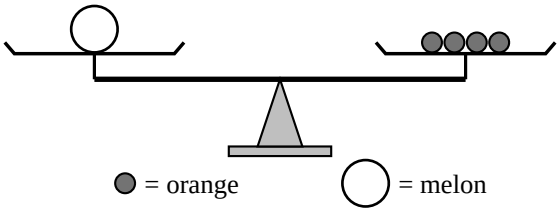
10. Mia cuts a square sheet of paper into 2 equal strips as shown. The perimeter of each strip is 48 cm.



How long is the side of the square?

A. 8 cm ***B.*** 13 cm ***C.*** 15 cm ***D.*** 16 cm ***E.*** 64 cm

11. The balance shows that one melon weighs as much as 4 oranges. 3 melons and 1 orange weigh 1625 g altogether.



How much does one orange weigh?

A. 100 g ***B.*** 125 g ***C.*** 250 g ***D.*** 500 g ***E.*** 1125 g

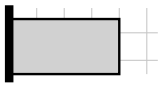
12. The odometer of a scooter shows 1479 km. Each digit is bigger than the digit before it. After how many more kilometres will this happen again?

A. 19 ***B.*** 10 ***C.*** 9 ***D.*** 8 ***E.*** 1

13. Nicolas, Omar, Ravi, Tom and Emma each have a different pet: a dog, a rabbit, a cat, a turtle and a parrot. Ravi has an animal without fur. Tom doesn't like cats. Emma has a pet that never barks. Tom has a pet that never barks. Nicolas has a pet that can fly.
Which statement is **not** true?

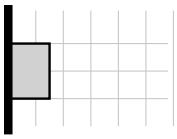
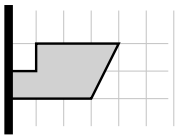
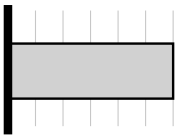
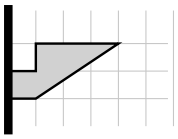
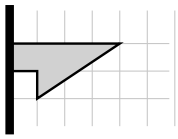
A. Nicolas has a parrot ***B.*** Emma has a turtle ***C.*** Emma has a cat
D. Omar has a dog ***E.*** Ravi has a turtle

14. From a window, Tom sees a rectangular flag on a pole. The wind turns and folds the flag, but the side sewn to the pole never moves. Tom draws the shadow of the flag on the wall at five different moments. Flat, the flag is 4 squares wide and 2 squares tall (see the picture). Four of the drawings are wrong.

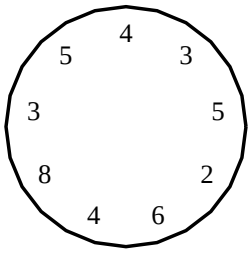


the flag, flat

Which drawing can be right?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

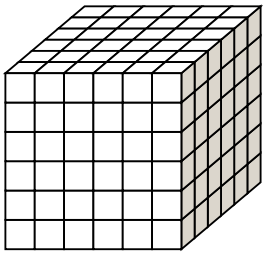
15. Numbers are written around a round plate as shown. Tom wants to break the plate with one straight line so that the two pieces have the same sum of numbers.



In how many different ways can he do it?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 5

16. Leo builds a brick 6 cubes long, 6 cubes deep and 6 cubes tall. To make it strong, the red outside is **two** cubes thick everywhere (also at the bottom, top, front and back). Everything inside that is blue.



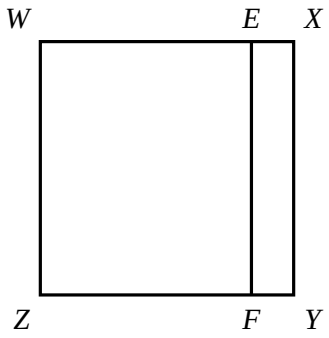
How many blue cubes are there?

A. 64 ***B.*** 27 ***C.*** 16 ***D.*** 8 ***E.*** 4

17. Five children, Clara, Anna, David, Hugo and Ben, sit in a row of 5 chairs at the cinema. Clara and Anna are best friends and want to sit next to each other.
In how many different ways can the children sit?

A. 72 ***B.*** 48 ***C.*** 24 ***D.*** 8 ***E.*** 4

18. The square *WXYZ* is cut into two rectangles. The perimeter of *WEFZ* is 22 cm and the perimeter of *EXYF* is 14 cm.



What is the perimeter of the square *WXYZ*?

A. 30 cm ***B.*** 24 cm ***C.*** 12 cm ***D.*** 9 cm ***E.*** 6 cm

19. The odometer of a scooter shows 1259 km. Each digit is bigger than the digit before it.
After how many more kilometres will this happen again?

A. 1 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

20. Fred has the cards 1, 4, 5 and 6. Using every card exactly once, with +, –, ×, ÷ and brackets, Fred wants to get the biggest possible number.
What is the biggest number Fred can get?

A. 16 ***B.*** 119 ***C.*** 120 ***D.*** 121 ***E.*** 150

Answer Key

1.	**A**	**2.**	**E**	**3.**	**D**	**4.**	**C**	**5.**	**E**	**6.**	**D**
7.	**C**	**8.**	**A**	**9.**	**D**	**10.**	**D**	**11.**	**B**	**12.**	**B**
13.	**B**	**14.**	**A**	**15.**	**D**	**16.**	**D**	**17.**	**B**	**18.**	**B**
19.	**E**	**20.**	**E**								

Solutions

1. A (24) — Clara's chair is fixed. The other 4 friends fill the other chairs: 4 × 3 × 2 × 1 = 24 ways. [2002 Q22 · fixed_seat]

2. E (Mia and Tom) — Fred: (1 + 9) × 4 = 40. Mia: with 5, 3 and 2 nothing gives 40 (the closest are 30 = 2 × 3 × 5); Tom: with 8, 7 and 6 nothing gives 40 (the closest are 34 = 6 × 7 – 8 and 41 = 6 × 8 – 7). [2002 Q21 · who_cannot]

3. D — All the other pieces can be found in the line, but this one cannot: it steps down where the line steps up. [2002 Q7 · cannot_belong]

4. C — Only this piece is found in the line. The others don't fit — A: it has two small bumps where the line has a pointed roof; B: it is a mirror image of a part of the line; D: it is a mirror image of a part of the line; E: it has a pointed roof where the line has a round dome. [2002 Q7 · can_belong]

5. E (84) — Count layer by layer, from the bottom: 7 × 7 + 5 × 5 + 3 × 3 + 1 × 1 = 84. [2002 Q20 · count_pyramid]

6. D (15) — 6 m = 600 cm. 600 ÷ 40 = 15. [2003 Q10 · pieces]

7. C (3) — The blue inside is 4 long and 2 tall, so one layer of it has 4 × 2 = 8 cubes. 8 ÷ 8 = 1 layers of blue, plus the red front and back: 3. [2003 Q9 · how_deep]

8. A (80) — The hole is 4 × 4 = 16 cubes in the front and goes 5 cubes back: 16 × 5 = 80. [2002 Q20 · fill_the_hole]

9. D (4 nuts) — Take 2 coconuts and 3 nuts off both plates. Left: 1 coconut = 4 nuts. [2002 Q9 · both_sides]

10. D (16 cm) — A strip is the full side long and 1/2 of it wide, so its perimeter is side + side + 2 × (side ÷ 2) = 6 × (side ÷ 2). 48 ÷ 6 = 8, so the side is 2 × 8 = 16 cm. [2002 Q6 · strips_to_square]

11. B (125 g) — Swap each melon for 4 oranges: 3 × 4 + 1 = 13 oranges weigh 1625 g. One orange: 1625 ÷ 13 = 125 g. [2002 Q9 · grams]

12. B (10) — Keep as many digits on the left as possible. The next such number is 1489: 1489 – 1479 = 10. [2002 Q24 · going_up]

13. B (Emma has a turtle) — The only possible matching: Nicolas – parrot; Omar – dog; Ravi – turtle; Tom – rabbit; Emma – cat. [2002 Q18 · which_not_true]

14. A — Check two things: the pole edge must be covered all the way, and nothing may reach more than 4 squares from the pole. Only picture A passes both checks. [2002 Q15 · only_one_right]

15. D (3) — Each half must have 20. The ways: 4+3+5+2+6 | 4+8+3+5; 3+5+2+6+4 | 8+3+5+4; 2+6+4+8 | 3+5+4+3+5. [2002 Q11 · count_ways]

16. D (8) — Two layers come off at each end of every length, so each length drops by 4: 2 × 2 × 2 = 8. [2003 Q9 · thick_wall]

17. B (48) — Think of Clara and Anna as one block: the block and the other 3 make 4 things to order, 4 × 3 × 2 × 1 = 24 ways. Inside the block Clara and Anna can swap, so 2 × 24 = 48. [2002 Q22 · side_by_side]

18. B (24 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 22 + 14 = 36 cm = 6 sides. Side = 36 ÷ 6 = 6 cm, perimeter 4 × 6 = 24 cm. [2002 Q6 · two_perimeters]

19. E (8) — Keep as many digits on the left as possible. The next such number is 1267: 1267 – 1259 = 8. [2002 Q24 · going_up]

20. E (150) — (1 + 4) × 5 × 6 = 150. Multiplying by 1 changes nothing, so it is better to add the 1 to the smallest other card first. [2002 Q21 · biggest]